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Renewable energy sources in the territory of east zangezur and Karabakh

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Abstract: Azerbaijan also has huge renewable energy resources. In particular, the solar energy potential of the liberated Azerbaijani lands is vast. This study examines the existing renewable energy resources of the liberated Azerbaijani lands, taking into account the underground and aboveground resources, without damaging the texture of the region. It should be noted that the construction of wind power plants in Kalbajar and Lachin, and solar power plants in Zangilan and Jabrayil, is planned. The renewable energy project in the Nagorno-Karabakh region serves to cover up Azerbaijan's recent ethnic cleansing of Armenians and its ecologically disastrous record. Now, Nagorno-Karabakh, whose entire civilian population has recently been subjected to ethnic cleansing, has been declared a "Green Energy Zone". A plan to transform the liberated lands into a renewable energy center has been announced and work has begun on this.

Keywords: renewable energy, Karabakh, Eastern Zangezur, solar energy, wind energy.

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1. Introduction

Azerbaijan is not only an energy-rich country, but also has great importance in terms of energy security, as it is a neighbor to energy-rich countries such as Kazakhstan, Turkmenistan, Russia and Iran. Azerbaijan's pipelines, railways and tanker ports in the Caspian Sea, and the infrastructure it has created have turned it into a transit country. Azerbaijan has transformed from a landlocked country into a transit country. Although some countries do not have energy resources, they can play the role of a bridge, thereby meeting their energy needs and increasing their strategic value as a transit country for buyer and seller countries. But of course, all this depends on creating a reliable image of the country.

Azerbaijan is also a country rich in renewable energy resources. Especially in 2020, the possibilities of producing electricity from renewable energy sources have increased with the lands liberated from occupation. This has become even more relevant in the modern world. Due to reasons such as climate change, drought and air pollution, countries are now preferring green and renewable energy sources. Obtaining energy from renewable energy sources has become of great importance in recent years. Now, not only countries with oil and natural gas, but also countries that can convert solar and wind energy into energy are energy-rich countries. Some countries even meet their electricity needs from wind, sun or water by selling their oil and natural gas. Some countries even export the electricity they get from renewable energy sources.

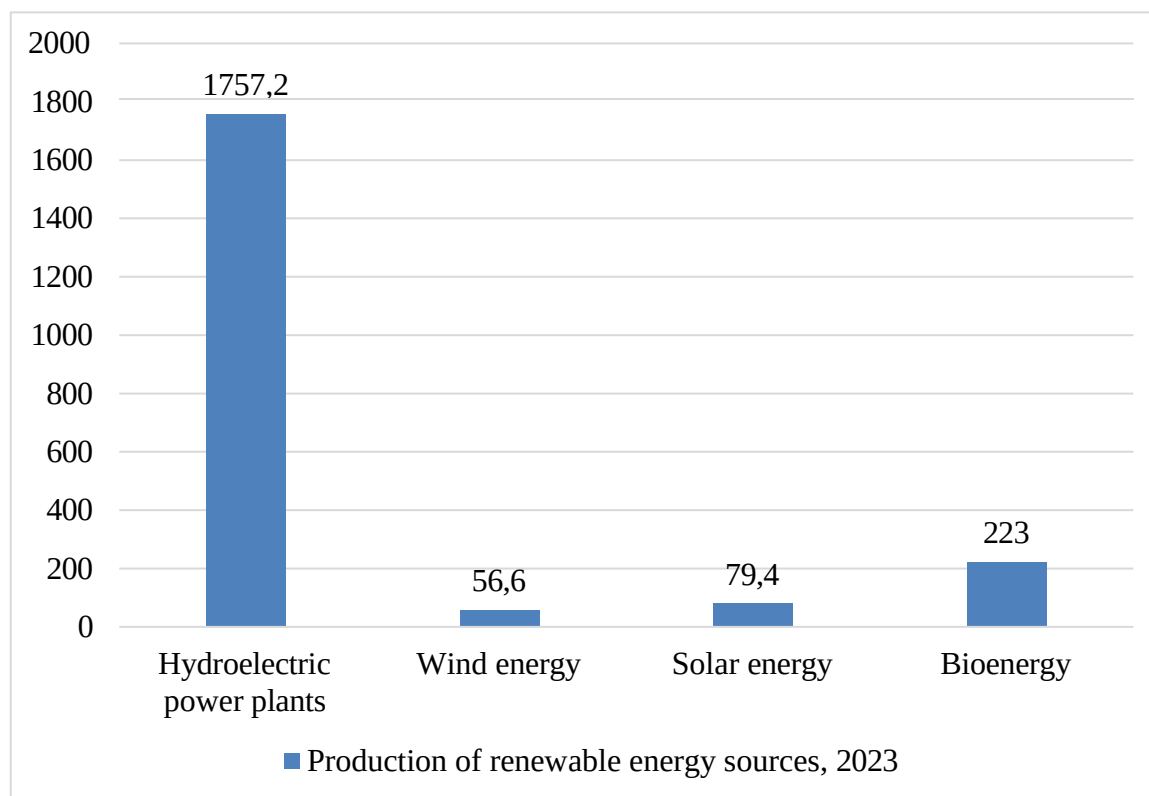
In modern times, Azerbaijan began to contribute to global energy security by signing the "Contract of the Century" in 1994 and continues to contribute to the pipelines it has repaired and built. Now, many neighboring countries prefer to export their energy resources through the infrastructure built by Azerbaijan. In this sense, it can be said that Azerbaijan is a reliable partner for both energy-rich countries and transit countries.

Although Azerbaijan is a hydrocarbon-rich country and is known worldwide in this field, in recent years Azerbaijan has paid attention to the purchase and use of energy from renewable energy sources, international cooperation in this field, capital investment and export of this energy.

According to the Ministry of Energy of Azerbaijan, the country's economic and technical renewable energy capacity is 27 thousand MW. Of this, 3000 MW is wind, 23,000 MW is solar, 380 MW is bioenergy and 520 MW is mountain rivers. Azerbaijan produces 17.3% of its total electricity production from renewable sources. 88.5% of renewable energy in the country is hydropower plants, 5.1% is wind, 3.5% is solar, and 2.9% is bioenergy sources [6].

2. Experimental detail

The lands liberated from occupation during the 44-day war are also an important powerhouse in terms of renewable energy sources. 25% of Azerbaijan's water resources are located in these lands. The solar potential in these lands is of particular importance in terms of electricity production. It is thought that the solar energy potential in the liberated lands exceeds 7,200 MW. In this sense, Azerbaijan has set itself the goal of increasing the power of renewable energy resources by signing contracts with various international companies. Azerbaijan has also set itself the goal of exporting natural gas, the main source of electricity production, by using the electricity it will produce from renewable energy sources within the country [3].



Graph 1. Electricity production at BOEM stations in 2023, million kWh Source: State Agency for Renewable Energy Sources under the Ministry of Energy of Azerbaijan, <https://area.gov.az/az#demo-363>

High technologies should be used to transform the areas under development in the Azerbaijani lands liberated from Armenian vandalism into "green energy" zones. The concepts of "smart city" and "smart village" are based on the development to be implemented. The creation of renewable electricity production sources is an important step for Azerbaijan in this sense. It is expected that the majority of renewable energy production in the region will be obtained from hydroelectric dams. It is no coincidence that Azerbaijan is turning to renewable energy sources and wants to use this technology in Karabakh, which it will rebuild. Although Azerbaijan, which ranks lowest among the South Caucasus countries in terms of green energy use, has a favorable geography for energy production, it uses natural gas reserves for electricity. For this reason, Azerbaijan has turned to renewable energy sources [5].

A quarter of Azerbaijan's internal water resources, that is, about 2.56 billion cubic meters of water per year, originates from Karabakh. The main rivers of the region, the Tartar, Bazarchay, Hakari and other small rivers, have great potential for hydropower. According to calculations, Karabakh has a solar energy potential of 3,000-4,000 megawatts, and a wind energy potential of 300-500 megawatts. The Khudafarin and Giz Galasi hydroelectric power stations, which are being built jointly with Iran, will produce 280 MW of energy, while the hydroelectric power stations planned for construction in the Kalbajar-Lachin region will produce 120 MW. This will allow the region to rely on green resources.

In addition, the thermal water resources of 3093 m³/day in the Kalbajar region and 412 m³/day in Shusha have great potential for the development of green energy. The main point here is not only to produce energy from green sources, but also to use the most efficient and intelligent systems in energy transmission, distribution and consumption [7].

The pilot village selected for the project is the Agali village of Zangilan. 200 houses to be built in the village will receive energy from renewable energy sources. Chinese Information and Communication Technologies (ICT) company Huawei is among the companies that will participate in smart village projects in the Karabakh region with

its advanced digital technologies. President Aliyev said at a press conference with Leyer that the liberated East Zangezur and Karabakh regions have an energy potential of 9200 megawatts. The wind energy potential in the Caspian Sea is 157 gigawatts" [8].

Azerbaijan will take advantage of renewable energy opportunities and use them for domestic consumption, thereby increasing natural gas exports to Europe. In addition, President Ilham Aliyev stated in his opening speech on Greek-Bulgarian relations on October 1, 2022 that Azerbaijan is negotiating with the participating states to export electricity produced from renewable energy sources to Europe. Azerbaijan, Turkey and Georgia have an agreement in the field of electricity. According to President Ilham Aliyev, a total of 1.6 billion kilowatt-hours of electricity were exported to Turkey, Russia, Georgia and Iran in 2021 [1].

Azerbaijan also exports electricity to Europe and Turkey through Georgia. Since May 1, 2019, Azerbaijan has been exporting electricity to European countries through Turkey and Georgia. The recipient countries in Europe are Bulgaria, Romania, Greece and Hungary. In the future, it plans to export electricity to Austria and Italy [4].

After the liberation of Azerbaijani lands from occupation, a second option for Azerbaijan's electricity export to Turkey and Europe emerged. Azerbaijan plans to export electricity to Turkey and Europe through the Zangezur corridor in the future.

3. Conclusion

Azerbaijan has also become an exporter of electricity. The country, which has been exporting electricity to Turkey and Georgia in recent years, plans to extend this power line to the Balkans in the future, and there are already demands for this. With the liberation of the territories from occupation in 2020, the country's renewable energy resources have increased. As a result, today Azerbaijan contributes to the world's energy security as both an energy source and a transit country. As an energy source, it is an exporter of oil, natural gas and electricity. As a transit country, it has become a reliable energy partner for both buyer and seller countries with its many pipeline options. Recently signed and renewed contracts indicate that its importance in the energy sector will increase further in the future.

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A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a cloudy sky. The image is framed by a dark green curved border at the top.

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A graphic of a water splash or wave, rendered in a soft, painterly style, located in the bottom right corner of the page.